Design a low-power **64-bit RISC-V compute subsystem** for **mobile XR**
under a **3 W, 3 nm-class LP envelope**,
returning a **DSE report with evidence and rejected alternatives**.

Intent
Improve mobile XR efficiency under power, reliability, memory, and deployment constraints.

Task
Bounded DSE for a RISC-V compute subsystem; start with an accelerator and memory slice.

Representation / world model
XRBench traces, architecture parameters, compiler assumptions, power model, constraints.

Environment / tool interface
Simulator or cost model plus workload harness; actions change tiling, memory, or vector width.

Method role
Generate candidates, predict cost, optimize, critique assumptions, and write evidence reports.

Feedback budget
Many cheap proxies, fewer simulations, rare high-fidelity checks before review.

Evidence
Pareto comparison over latency, energy, area proxy, memory traffic, QoS, and sensitivity.

Negative traces
Violates 3 W, misses real-time, exceeds bandwidth, unsupported software or fidelity failure.

Rejection authority
Checks, tool failures, QoS misses, review.

Human decision
Refine, escalate fidelity, change scope, or reject.

Example card, not a completed design claim.